

FIFA World Cup 2026 Tournament Network

VLSM Addressing Plan

Student ID used for this addressing plan: 22131547

Base Network: 15.47.0.0/16

1. LAN VLSM Allocation

Headquarters	Required Hosts	Network Address	Prefix	Subnet Mask	First Usable	Last Usable	Broadcast
Brazil	280	15.47.0.0	/23	255.255.254.0	15.47.0.1	15.47.1.254	15.47.1.255
Argentina	210	15.47.2.0	/24	255.255.255.0	15.47.2.1	15.47.2.254	15.47.2.255
France	170	15.47.3.0	/24	255.255.255.0	15.47.3.1	15.47.3.254	15.47.3.255
Spain	90	15.47.4.0	/25	255.255.255.128	15.47.4.1	15.47.4.126	15.47.4.127
England	130	15.47.6.0	/24	255.255.255.0	15.47.6.1	15.47.6.254	15.47.6.255
Germany	60	15.47.5.128	/26	255.255.255.192	15.47.5.129	15.47.5.190	15.47.5.191

2. Point-to-Point WAN VLSM Allocation

Link	Network	Prefix	Subnet Mask	Router 1	Router 2
France–Spain	15.47.7.0	/30	255.255.255.252	France: 15.47.7.1	Spain: 15.47.7.2
France–England	15.47.7.4	/30	255.255.255.252	France: 15.47.7.5	England: 15.47.7.6
France–Germany	15.47.7.8	/30	255.255.255.252	France: 15.47.7.9	Germany: 15.47.7.10
England–Germany	15.47.7.12	/30	255.255.255.252	England: 15.47.7.13	Germany: 15.47.7.14

3. Shared Brazil–Argentina–France Layer-2 Network

Network	Prefix	Subnet Mask	Device	IP Address
15.47.7.16	/29	255.255.255.248	Brazil Router	15.47.7.17
15.47.7.16	/29	255.255.255.248	Argentina Router	15.47.7.18
15.47.7.16	/29	255.255.255.248	France Router	15.47.7.19

4. Key Device IP Addresses

Device	Location	IP Address
Brazil Router LAN Gateway	Brazil	15.47.0.1
Argentina Router LAN Gateway	Argentina	15.47.2.1
France Router LAN Gateway	France	15.47.3.1
Spain Router LAN Gateway	Spain	15.47.4.1
England Router LAN Gateway	England	15.47.6.1
Germany Router LAN Gateway	Germany	15.47.5.129
Brazil Web Server	Brazil	15.47.0.20
France DNS Server	France	15.47.3.20
Germany Web Server	Germany	15.47.5.150

5. DNS Records

Domain Name	Server IP	Record Type
www.brazil2026.com	15.47.0.20	A Record
www.germany2026.com	15.47.5.150	A Record

6. VLSM Ordering

The LANs are allocated from the largest host requirement to the smallest. Brazil requires 280 hosts, followed by Argentina (210), France (170), England (130), Spain (90), and Germany (60). Point-to-point links use /30 subnets, while the shared Brazil–Argentina–France Layer-2 network uses /29.